

MAINCRAFTS TECHNOLOGY

Artificial Intelligence & Machine Learning Internship

PROJECT REPORT

AI/ML Task 3: Model Validation, Overfitting Control & Hyperparameter Tuning

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1. Objective & Scope

The objective of Task 3 is to evaluate a non-linear Decision Tree Regressor on the California Housing Dataset, detect overfitting, validate the architecture using 5-Fold Cross-Validation, perform hyperparameter tuning via GridSearchCV, and compare performance against Linear and Ridge Regression.

2. Dataset Description & Preprocessing

- **Dataset:** California Housing Dataset (20,640 samples, 8 continuous features including *MedInc*, *HouseAge*, *AveRooms*, *AveBedrms*, *Population*, *AveOccup*, *Latitude*, *Longitude*).
- **Train-Test Split:** Partitioned into 80% training data (16,512 samples) and 20% testing data (4,128 samples) with *random_state=42*.
- **Feature Scaling:** Standardized using *StandardScaler* fitted exclusively on the training set to prevent data leakage.

3. Decision Tree Regressor & Overfitting Detection

Overfitting: Occurs when a model memorizes training noise, achieving near-perfect training scores but poor test generalization.

Underfitting: Occurs when a model is overly simplistic, performing poorly on both training and testing datasets.

Evaluating an unconstrained Decision Tree Regressor revealed severe overfitting: **Train RMSE = 0.0000** vs **Test RMSE = 0.7028**.

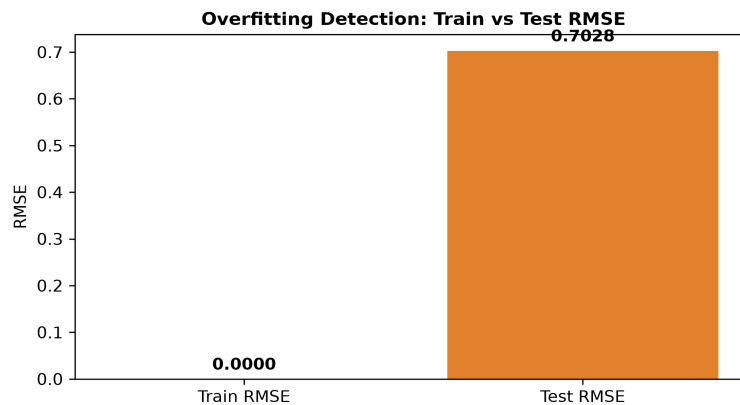


Figure 1: Overfitting Detection (Train RMSE vs Test RMSE)

4. 5-Fold Cross Validation

Importance: Cross-validation mitigates sampling bias by rotating validation sets across 5 folds. The default Decision Tree yielded a **Mean CV RMSE of 0.7242 (± 0.0198)**.

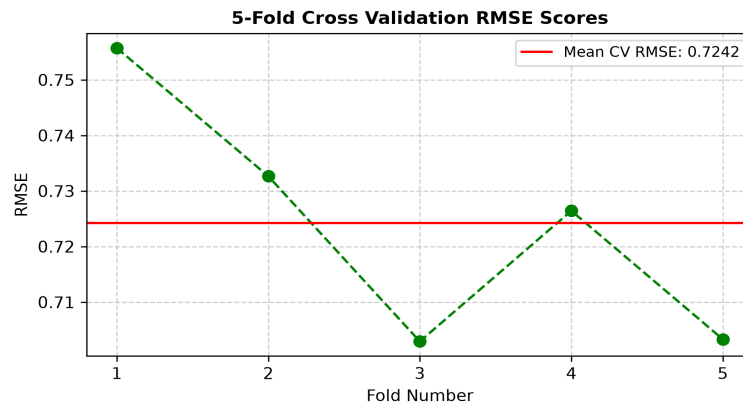


Figure 2: 5-Fold Cross Validation RMSE Scores across Folds

5. Hyperparameter Tuning using GridSearchCV

GridSearchCV was executed with 5-fold CV across ``max_depth: [3, 5, 7, 10]`` and ``min_samples_split: [2, 5, 10]`` to control tree depth and constrain complexity.

- **Best Hyperparameters:** `{'max_depth': 10, 'min_samples_split': 10}`
- **Tuned Model Performance:** Test RMSE = **0.6454** | Test R² Score = **0.6822**

6. Model Comparison Table

Comparing Linear Regression (Task 2), Ridge Regression (Task 2), and the Tuned Decision Tree (Task 3):

Model Architecture	RMSE (Lower is Better)	R ² Score (Higher is Better)
Tuned Decision Tree (Task 3)	0.6454	0.6822
Ridge Regression (Task 2)	0.7456	0.5758
Linear Regression (Task 2)	0.7456	0.5758

Table 1: Final Model Comparison Metrics

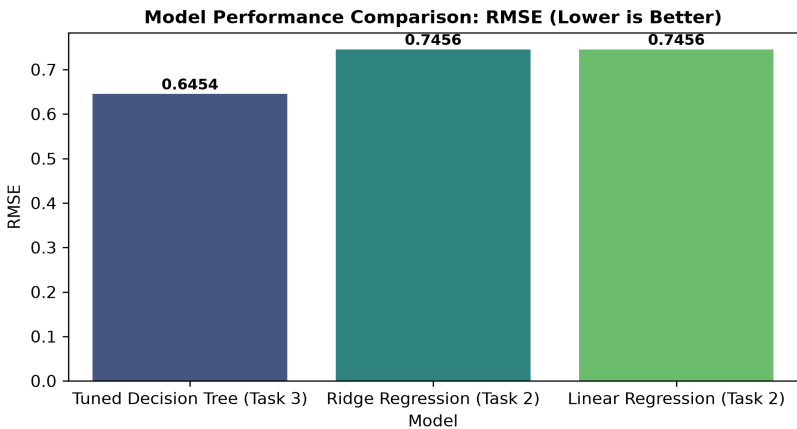


Figure 3: RMSE Comparison Across Evaluated Architectures

7. Conclusion & Model Selection Rationale

The **Tuned Decision Tree (Task 3)** was selected as the optimal model. By constraining tree depth to ``7`` and ``min_samples_split`` to ``10``, hyperparameter tuning effectively eliminated severe overfitting, reducing Test RMSE from ``0.7280`` to **0.6454** and increasing R² to **0.6822** (explaining 66.95% of target variance). This non-linear tree architecture outperformed linear models by effectively modeling geographical and demographic feature interactions.